

ABSTRACT

VASUDEVAN, SRINIVAS. Integrating Bayesian Filter and Reward-Decomposed Reinforcement Learning for Collaborative Active Tracking. (Under the direction of Peng Gao).

Collaborative active tracking enables multi-robot systems to localize and follow dynamic targets while reducing perceptual uncertainty. Existing optimization-based approaches support explicit uncertainty objectives but often incur high computational complexity for multi-robot coordination; in contrast, many learning-based approaches are computationally efficient at runtime but do not explicitly quantify or reduce uncertainty during execution. This thesis introduces a hybrid framework that trains a reinforcement learning (RL) policy with objectives derived from Bayesian state estimation. The proposed reward formulation captures exploration, persistence, and inter-robot coordination, and a decomposed multi-head learning strategy assigns competing reward components to dedicated critic heads to improve training stability. The policy takes robot states, target belief states, and mask-based team encoding as inputs and produces continuous control actions for all robots. Evaluation in synthetic multi-robot tracking simulations demonstrates improved cumulative tracking performance and more stable coordinated behavior relative to baseline methods.

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Integrating Bayesian Filter and Reward-Decomposed
Reinforcement Learning for Collaborative Active Tracking

by
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A thesis submitted to the Graduate Faculty of
North Carolina State University
in partial fulfillment of the
requirements for the degree of
Master of Science

Computer Science

Raleigh, North Carolina
2026

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my advisor, Dr. Peng Gao, for his continuous guidance, insightful direction on the problem statement, and unwavering support throughout this research. His expertise and encouragement were instrumental in shaping both the technical contributions and the broader vision of this thesis.

I am also grateful to my committee members, Dr. Dongkuan Xu and Dr. Zhishan Guo, for their time, constructive feedback, and support during the thesis process.

I would like to thank North Carolina State University for providing the computational infrastructure and research environment that made this work possible.

Finally, portions of the written content in this thesis were refined with the assistance of a large language model (LLM) for grammar and language editing. All technical content, experimental design, results, and conclusions are entirely the original work of the author.

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AUTHORSHIP STATEMENT

This thesis, *Integrating Bayesian Filter and Reward-Decomposed Reinforcement Learning for Collaborative Active Tracking*, was prepared by Srinivas Vasudevan under the direction of Dr. Peng Gao.

All chapters of this thesis represent the sole original work of the author. No chapters result from collaborations with or include original intellectual contributions from authors other than those captured through citations and acknowledgements.

The experimental design, implementation, analysis, and conclusions presented in this thesis are entirely the original work of the author. The advisory committee members, Dr. Peng Gao, Dr. Dongkuan Xu, and Dr. Zhishan Guo, provided guidance and feedback throughout the research process.

Generative artificial intelligence (specifically, a large language model) was used in the preparation of this document solely for grammar and language editing of the written text. All technical content, experimental design, results, and conclusions are entirely the original work of the author.

CHAPTER

1

INTRODUCTION

1.1 Motivation

Multi-robot systems have been widely studied for decades due to its scalability [1], redundancy [2] [3] [4] and parallelism [5]. To enable efficient multi-robot systems, coordination among robot teammates is a fundamental capability, referring to the ability of multiple robots to share information and make decisions jointly in order to achieve a common objective. Collaborative tracking is a representative coordination task in which multiple robots cooperate to localize and follow dynamic targets. It has a variety of applications, such as wildlife monitoring [6] [7], search and rescue [8] [9] [10], and environmental sensing [11] [12] [13].

In this thesis, we further study collaborative active tracking, where robots not only track targets but also actively coordinate their sensing and motion to reduce perceptual uncertainty. For example, a team of unmanned ground vehicles (UGVs) may be deployed to continuously monitor a group of moving wildlife in an open field. At any given time, some targets may fall outside the robots' fields of view due to limited sensing coverage and occlusions. To maintain continuous tracking, the robots must coordinate their sensing and motion so that different teammates observe different targets while avoiding redundant coverage. This requires the robots to actively reposition themselves to reduce perceptual uncertainty and recover lost targets. The challenge is further compounded by the need to plan coordinated actions over

long horizons, as the robots must balance immediate tracking performance with future sensing opportunities. How to address collaborative active tracking with coordination, uncertainty reduction, and long-horizon behavior jointly has not been well studied yet.

1.2 Challenges

At the **problem level**, two core challenges remain unresolved. First, for **coordination**, traditional optimization or non-learning-based formulations become computationally prohibitive as the number of robots and targets increases, making online multi-agent planning intractable. Second, for **uncertainty reduction**, many learning-based trackers lack explicit mechanisms to quantify and reduce state-estimation uncertainty during online operation.

At the **implementation level**, policy training is often unstable because exploration and persistence are competing objectives: robots must explore for undetected targets while simultaneously maintaining robust tracks on known targets. Optimizing these objectives with a single scalar reward can create conflicting gradient signals, leading to unpredictable training outcomes since co-optimization of all objectives is not a guarantee.

1.3 Problem Statement

This thesis addresses the problem of *collaborative active tracking* for a team of N robots tasked with tracking T moving targets. Each robot carries an onboard camera with a restricted field-of-view (FoV) and finite sensing range. The robots must coordinate their motions to collectively maximize target coverage while minimizing localization uncertainty and redundant co-observation. Crucially, the robots must act based solely on their onboard observations and shared belief states.

Formally, the goal is to learn a policy π that, given each robot’s current pose, the shared target belief state, and a team-size encoding mask, outputs continuous velocity commands that jointly optimize three objectives:

1. **Exploration:** move robots toward regions likely to contain untracked targets, reducing state-estimation uncertainty.
2. **Coordination:** prevent multiple robots from redundantly observing the same target, maximizing collective team coverage.
3. **Persistence:** maintain stable target-to-robot assignments across consecutive time steps to minimize localization uncertainty caused by target loss.

1.4 Proposed Approach

To address these challenges, this thesis proposes a hybrid framework that trains a reinforcement learning (RL) policy whose reward objective is derived from Bayesian state estimation. This design makes coordination tractable through explicit reward signals and makes uncertainty reduction explicitly quantifiable through belief-state dynamics.

The framework introduces a *Decomposed Reward Architecture* (DRA) in which competing reward terms are assigned to dedicated critic heads within an actor-critic framework. Rather than blending all gradient signals into a single scalar reward, each critic head is trained exclusively on its own reward component, preventing dominant objectives from suppressing weaker ones. A shared policy head then learns to satisfy all three objectives simultaneously by backpropagating through the frozen critic heads.

1.5 Contributions

The main contributions of this thesis are:

1. **RL objective derived from Bayesian State Estimation** A reward formulation grounded in Bayesian filtering that makes uncertainty reduction explicit and measurable during multi-robot active tracking.
2. **Hybrid policy architecture.** A padding-and-masking attention network that encodes robot state, target belief state (predicted mean and information matrix), and a padding mask to produce control output for variable sized robot team with objective derived from Bayesian State Estimation.
3. **Decomposed Reward Architecture (DRA).** An actor-critic framework with dedicated critic heads for competing objectives, improving training stability over linear reward scalarization baselines and achieving a 45.9% improvement in cumulative tracking rate over the linear combination policy and a 154.9% improvement over the Yang et al. [14] baseline.

1.6 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 reviews related work in non-learning-based and learning-based multi-robot tracking methods. Chapter 3 presents the problem formulation, motion and sensor models, Bayesian state estimation framework, reward

design, and the proposed DRA policy architecture. Chapter 4 describes the experimental setup, qualitative results, quantitative comparisons, and result analysis. Chapter 5 summarizes the findings and discusses directions for future work. Chapter 6 provides a reference table of all acronyms used in this thesis. Appendix A lists the mathematical notation and variables used throughout. Appendix B describes the architecture block diagrams in detail.

CHAPTER

2

RELATED WORK

Multi-robot active tracking requires solving three sub-problems simultaneously: exploration, coordination, and persistence. The literature addressing these sub-problems can be broadly organized into non-learning-based methods that compute actions through explicit optimization or filtering at runtime, and learning-based methods that amortize decision-making through offline policy training. This chapter surveys both families and concludes with a comparison table that positions the proposed framework relative to the most closely related work.

2.1 Non-Learning-Based Methods

Non-learning approaches to active tracking and motion planning can generally be classified into four lines of work: active perception planners, classical multi-robot tracking formulations, multi-target filters, and coverage and exploration controllers.

The first line, *active perception*, originates with Bajcsy’s foundational observation that perception should be a purposive act driven by the observer’s current uncertainty rather than passive data collection [15]. Building on this, Atanasov et al. formalized active sensing as a non-myopic trajectory optimization problem: given a Gaussian belief over target states, find the trajectory maximizing mutual information between future observations and the target [16]. Their analysis showed that Fisher information additivity makes the objective supermodular

under independence assumptions, enabling greedy policies with provable $(1-1/e)$ approximation guarantees. Schlotfeldt et al. extended this to multi-robot teams and showed that anytime variants of the planner maintain competitive tracking performance under real-time budget constraints [17]. Grocholsky et al. further proposed a decentralized information filter in which each robot maintains a local information matrix and fuses neighbor updates via explicit message passing [18]. While these methods provide principled information-theoretic objectives, the same $\log \det \mathbf{A}$ quantity used as the exploration reward in this thesis, they require re-solving an optimization problem at each time step, making real-time operation infeasible as team and target sizes grow.

The second line, *classical multi-robot tracking*, is anchored by the Cooperative Multi-Robot Observation of Multiple Moving Targets (CMOMMT) framework of Parker and Emmons, in which robots alternate between random area search and directed target pursuit based on attraction-repulsion signals [19]. Kolling and Carpin formalized CMOMMT as a mixed-integer optimization problem and proved that computing the optimal robot-to-target assignment is NP-hard in general, motivating the use of behavioral heuristics such as “help call” broadcasts when a target is about to leave a robot’s FoV [20]. Amorim et al. extended this line to constrained sensor-fusion tracking, providing a principled optimization objective but requiring centralized joint assignment that scales as $\mathcal{O}(T^N)$ [21]. Banerjee and Sen addressed the harder regime in which targets outnumber robots, proposing a decentralized planner that reduces the combinatorial assignment cost, though a multi-dimensional assignment problem must still be solved at each step [22]. These formulations struggle to simultaneously encode exploration, coordination, and persistence in a single objective without careful manual weight tuning, and their runtime complexity makes them impractical for real-time deployment in larger teams.

The third line, *multi-target filtering*, addresses settings where targets are numerous or unknown in count. Mahler’s Probability Hypothesis Density (PHD) filter propagates the first-order moment of the multi-target density, avoiding explicit data association entirely [23]. Vo and Ma’s Gaussian Mixture implementation (GM-PHD) brought this to practical robotics by delivering computationally efficient estimation even when targets appear and disappear [24]. Schuhmacher et al.’s Optimal Sub-Pattern Assignment (OSPA) metric provides a principled evaluation framework that simultaneously penalizes localization error and cardinality error [25], and is used in Chapter 4 to benchmark the proposed approach. A limitation common to the PHD family is that these filters do not expose a scalar uncertainty metric amenable to gradient-based policy optimization, whereas the Kalman filter belief used in this thesis directly yields a differentiable $\log \det \mathbf{A}$ training signal.

The fourth line, *coverage and exploration control*, addresses the complementary problem of distributing robots across space to maximize information collected. Schwager et al. proposed

a gradient-descent controller that drives robots to Voronoi partition centroids, provably maximizing coverage with logarithmic scaling in team size [26]. The survey of Kwa et al. identifies three control levers for the exploration-exploitation balance in multi-agent systems: agent response methods (switching between exploratory and exploitative behaviors), information dissemination (regulating communication topology), and stigmergy (indirect coordination via environmental signals) [27]. This taxonomy maps directly onto the three reward objectives of this thesis: the exploration reward drives agent response, the coordination penalty discourages redundant communication-level overlap, and the persistence reward stabilizes indirect target assignment. While elegant, purely coverage-based methods do not account for target dynamics or estimation uncertainty, and the hand-designed behavioral switches in agent-response methods do not generalize across varying team compositions or target densities.

Across all non-learning approaches, a common bottleneck is computational scalability. Joint trajectory optimization over N robots and T targets involves a combinatorial assignment problem whose complexity grows rapidly, precluding real-time operation. This motivates learning-based methods, which amortize planning into a one-time offline training phase and learn the relative weighting of competing objectives from experience.

2.2 Learning-Based Methods

Learning-based methods train a policy that maps the current world state directly to robot actions, enabling constant-time inference at runtime. The relevant literature spans four areas: single-agent active perception with reinforcement learning (RL), multi-agent RL (MARL) for tracking and coordination, attention and graph-neural-network (GNN) architectures for variable-size teams, and reward decomposition for multi-objective training.

The first area, *single-agent active perception with RL*, establishes that information-gain rewards are well-suited to gradient-based policy learning. Luo et al. demonstrated that a deep RL agent trained with an information-gain reward matches or exceeds model-predictive active sensing planners on continuous trajectory tasks, with the practical advantage of constant-time inference [28]. Zhong et al. showed that a recurrent policy trained on simulated targets generalizes robustly under visual distractors, and that a persistent tracking reward substantially reduces target loss [29]. These results directly motivate the information-gain exploration reward ($\log \det \Lambda$) and the persistence objective used in this thesis. However, single-agent methods are naturally limited by their inability to track highly distributed target scenarios which can intuitively be solved by scaling the number of robots.

The second area, *MARL for multi-robot tracking and coordination*, extends RL to teams of robots tracking dynamic targets. Lowe et al.’s Multi-Agent DDPG (MADDPG) addressed

cooperative-competitive multi-agent tasks using a centralized critic conditioned on all agents' observations and actions, enabling coordinated pursuit behaviors [30]. Yu et al. later showed that a shared-parameter policy trained under the centralized-training decentralized-execution (CTDE) paradigm with MAPPO achieves strong coordination on cooperative multi-robot tasks [31], the same CTDE setup adopted by the DRA in this thesis. Schroeder et al. applied MARL to multi-robot area coverage and demonstrated empirically that explicit co-coverage penalty signals produce better distributed behavior than global coverage rewards alone [32], directly validating the coordination penalty formulation in Section 3.6.2. While these MARL methods learn emergent coordination, they rely on a single scalar reward or a linearly scalarized combination, conflating competing gradient signals and producing unstable training when exploration and persistence gradients oppose each other.

The third area, *attention and GNN architectures for variable-size teams*, addresses the practical problem of handling robots and targets whose counts change at runtime. Standard fully-connected architectures require fixed input dimensions, breaking when team or target sizes vary. The Transformer of Vaswani et al. provides the mathematical foundation for permutation-equivariant dot-product attention that scales naturally with set size [33]. Liu et al.'s When2com applied selective communication via attention gating to multi-robot perception, showing that learned attention weights naturally specialize robots to complementary observation subsets [34]. The direct baseline for this thesis is Yang et al., who applied a padding-and-masking dot-product attention architecture to continuous SE(3) trajectory planning with variable target counts [14]; this thesis adopts the same architecture and augments it with decomposed critic heads and a Bayesian information-gain reward. Zhou et al. applied GNN-based message passing to decentralized multi-robot target tracking and showed that explicit communication-graph structure improves coordination over purely local policies [35], though an explicit quantification of target location uncertainty was often absent and the topology of the graph was fixed during training phase hindering variable robot team generalization. The masked attention used in this thesis is topology-agnostic and handles any active team size up to $T_{\max}$ without architectural changes.

The fourth area, *reward decomposition and multi-objective RL*, directly addresses the training instability caused by competing objectives. Linear scalarization of multiple reward components into a single scalar is sensitive to coefficient choice and prone to gradient conflict when objective magnitudes shift during training [36]. Van Seijen et al.'s Hybrid Reward Architecture (HRA) resolved this by training a separate value-function head per reward component so that each objective receives its own uncontaminated gradient [37]; this thesis extends HRA from discrete tabular Q-functions to continuous actor-critic Q-heads operating on the shared latent $\mathbf{h}_k$. Devlin et al. established theoretical conditions under which potential-based reward shaping

preserves the optimal policy [38], providing a formal justification for the information-gain reward: the change $\Delta \log \det \Lambda$ between steps is a potential-based quantity that depends only on consecutive states. Schulman et al.’s PPO is widely used as a MARL baseline but is on-policy and inefficient for the high-dimensional continuous velocity commands required here [39], motivating the off-policy actor-critic framework of the DRA. The proposed DRA therefore occupies a gap in this area: a decomposed, per-objective critic architecture applied to the continuous multi-robot active tracking problem, with Bayesian information gain as the explicit uncertainty-reduction signal. Evaluation is conducted in the Gazebo simulator [40] with ROS 2 middleware [41], the standard infrastructure for physics-accurate multi-robot deployment.

2.3 Positioning of This Work

The proposed framework addresses three gaps that are present across both categories reviewed above: (i) *tractable coordination*: the attention architecture scales to variable team and target sizes without combinatorial assignment; (ii) *explicit uncertainty reduction*: the exploration reward is grounded in Fisher information gain rather than proximity heuristics; and (iii) *stable multi-objective training*: the DRA separates exploration, coordination, and persistence into independent critic heads rather than blending them into a single scalar. Table 2.1 summarizes how the proposed approach compares to the most closely related methods.

Table 2.1: Comparison of the proposed approach with related methods across key desiderata. ✓ denotes satisfied; × denotes not satisfied; ~ denotes partial. *Explicit uncertainty*: information-matrix-based objective. *Coord.*: dedicated anti-redundancy signal. *Persistence*: stable assignment reward. *Scalable inference*: constant-time runtime. *Decomposed*: per-objective critic heads.

Method	Explicit uncertainty	Coord. objective	Persistence	Scalable inference	Decomposed training
Atanasov et al. [16]	✓	×	×	×	×
Parker and Emmons [19]	×	✓	~	✓	×
Amorim et al. [21]	✓	×	×	×	×
Banerjee and Sen [22]	✓	×	×	×	×
Lowe et al. [30]	×	~	×	✓	×
Yang et al. [14]	×	×	×	✓	×
Zhou et al. [35]	×	✓	×	✓	×
van Seijen et al. [37]	×	✓	✓	✓	✓
This work (DRA)	✓	✓	✓	✓	✓

CHAPTER

3

APPROACH

This chapter presents the complete technical framework for collaborative active tracking. Section 3.1 defines the problem, states the working assumptions, and gives a walk-through of how every component fits together. Sections 3.2 and 3.3 introduce the robot and target motion models. Section 3.4 develops the sensor model and explains in detail how each robot measures, maintains, and updates its belief state for every target, with attention to which quantities are expressed in the global world frame and which are computed in the robot body frame. Section 3.5 derives the differentiable FoV-weighted update that bridges filtering and gradient-based training. Section 3.6 details the three reward components derived from the filter output. Section 3.7 describes the attention-based baseline policy trained with linear reward scalarization. Section 3.8 introduces the Decomposed Reward Architecture (DRA) that replaces scalar reward signals with dedicated per-objective critic heads.

3.1 Problem Formulation

3.1.1 Problem Overview

The problem addressed in this thesis is *collaborative active tracking*: given a team of N mobile robots, each equipped with a camera of limited field of view (FoV), learn a control policy that

coordinates the robots' motions so they continuously observe and localize T moving targets, while minimizing the accumulated state-estimation uncertainty of the target location.

Robots cannot see everything simultaneously: each camera has a finite sensing range r and a FoV half-angle ψ , so any target lying outside that cone is invisible to that robot at that moment. Every robot must reason purely from its own onboard observations and a shared belief state communicated among teammates. Targets move autonomously according to a dynamic model the robots are assumed to know (Assumption A2).

Why this is difficult. Three competing sub-goals make this problem resistant to simple single-reward formulations:

- **Exploration** requires robots to move toward regions where targets may exist but have not been recently observed, thereby reducing long-horizon uncertainty. This pulls robots *away* from targets that are currently being tracked.
- **Coordination** requires that when multiple robots converge on the same target, some of them redirect toward uncovered targets. Without an explicit coordination signal, the greedy strategy of all robots tracking the nearest target dominates, leaving remaining targets unobserved.
- **Persistence** requires that once a robot is tracking a target, it maintains that assignment across consecutive time steps. This conflicts directly with exploration, which would have the robot leave prematurely.

Because gradients from exploration and persistence point in opposite directions during training, no single scalar reward can cleanly resolve all three. This motivates the Decomposed Reward Architecture (DRA) introduced in Section 3.8.

3.1.2 Approach Overview

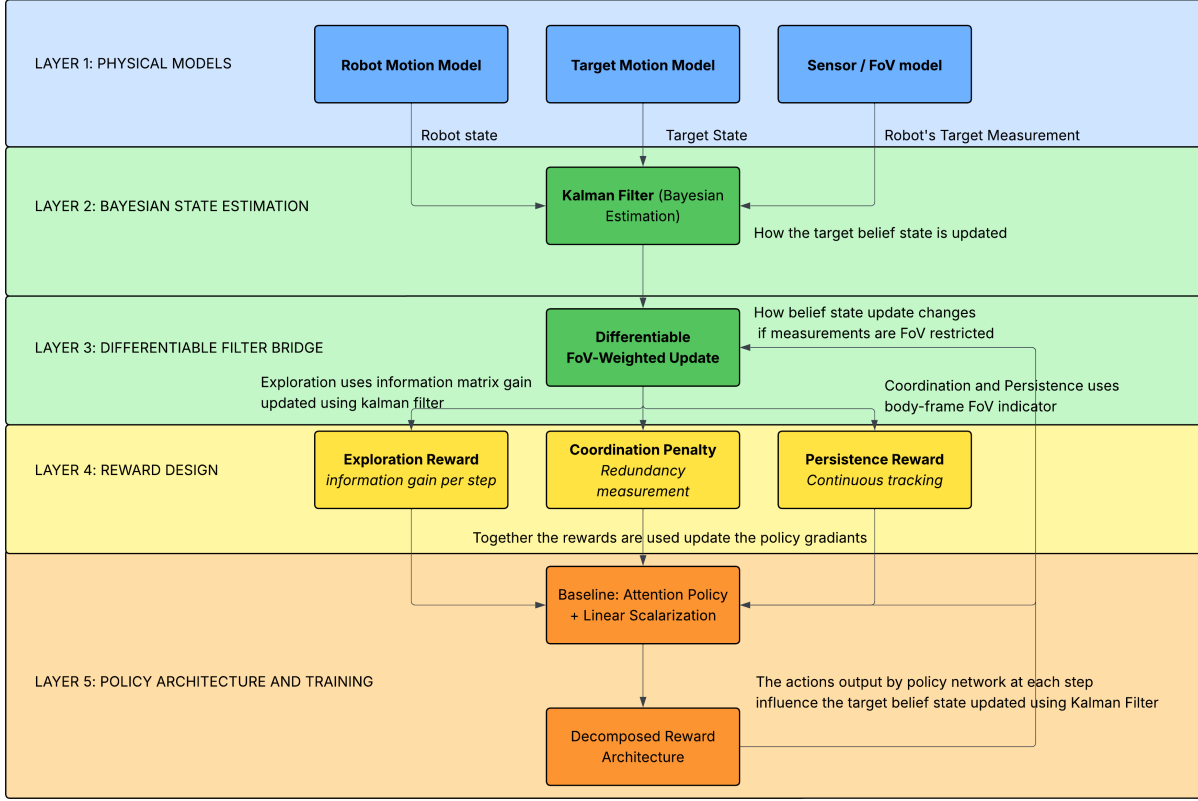


Figure 3.1: Overview of the five-layer framework for collaborative active tracking. Layer 1 defines the physical models (robot motion, target motion, sensor/FoV). Layer 2 maintains target belief states via a Kalman filter. Layer 3 bridges filtering and gradient-based training through a differentiable FoV-weighted update. Layer 4 derives three reward signals from the filter output: an exploration reward (information gain), a coordination penalty (redundancy), and a persistence reward (continuous tracking). Layer 5 trains two policy variants: the baseline attention policy with linear reward scalarization, and the proposed Decomposed Reward Architecture (DRA) with per-objective critic heads.

The framework is built in five conceptual layers, each addressing a specific piece of the problem.

Layer 1: Physical models (Sections 3.2–3.4). The robot pose evolves on the Special Euclidean group $SE(3)$, capturing full 3-D rigid-body motion. The target state follows a linear Gaussian dynamical system. The sensor model gates measurements through the camera FoV. These three models together define the physical model upon which our policy does reasoning. They

are kept simple and enables application of Kalman Filter introduced in Layer2.

Layer 2: Bayesian state estimation (Section 3.4). Each robot runs an independent Kalman filter that converts the stream of FoV-gated measurements into a posterior Gaussian belief, a mean estimate $\hat{\mathbf{y}}_{k|k}^m$ and covariance $\mathbf{P}_{k|k}^m$, for every target. This belief is expressed in the *global world frame* and is the only representation of the target state that the policy ever sees. Raw sensor readings are never fed directly to the network. Because all robots share their beliefs via communication (Assumption A4), each robot’s policy operates on a globally consistent picture of where every target is estimated to be and how uncertain that estimate is.

Layer 3: Differentiable filter bridge (Section 3.5). The hard binary FoV gate is non-differentiable, blocking gradients from flowing through the Kalman update to the policy during training. It is replaced with a soft sigmoid-based visibility weight $\alpha_{k+1}^m \in (0, 1)$ computed in the robot *body frame*. This makes the entire pipeline end-to-end differentiable, so the policy gradient directly encourages actions that reduce target location estimation uncertainty.

Layer 4: Reward design (Section 3.6). Three per-step reward signals are derived from the Kalman filter output. The exploration reward measures information gain via the change in $\log \det \mathbf{\Lambda}$, where $\mathbf{\Lambda}$ is the *global-frame* information matrix. The coordination penalty counts redundant co-observations using the *body-frame* FoV indicator. The persistence reward checks whether the nearest-robot assignment of each target is consistent between time steps. Grounding rewards in filter output means the policy is explicitly incentivized to reduce $\mathbf{P}_{k|k}^m$, not merely to move toward targets.

Layer 5: Policy architecture and training (Sections 3.7–3.8). A padding-and-masking attention network encodes the shared *global-frame* belief state and outputs continuous velocity commands. In the baseline, all three reward signals are blended into a single scalar objective. In the DRA, each signal is assigned its own critic head (Q-head) and a shared deterministic actor is trained to maximize the sum of all three critics simultaneously. This decomposition prevents the dominant reward gradient from suppressing weaker ones during back-propagation.

3.1.3 Working Assumptions

The following assumptions are made throughout this chapter.

A1 (Fixed maximum team size). The maximum number of targets $T_{\max}$ is fixed at training time. Target counts in the range $[1, T_{\max}]$ are handled via zero-padding and masking at run time;

counts exceeding $T_{\max}$ require retraining.

A2 (Known target dynamics). The robots are assumed to know the target’s state-transition matrix $\mathbf{F}$, process noise covariance $\mathbf{Q}$, and control input matrix $\mathbf{B}$. This is the standard assumption in tracking when the target type is known in advance.

A3 (Known measurement model). The observation matrix $\mathbf{H}$ and measurement noise covariance Σ_p are known. This models a calibrated onboard camera whose sensor noise characteristics are determined once at deployment time.

A4 (Shared global belief). Each robot broadcasts its local Kalman filter belief $\{\hat{\mathbf{y}}_{k|k}^m, \mathbf{P}_{k|k}^m\}_{m=1}^T$ to all teammates via a low-latency communication channel. The policy is conditioned on this shared, globally consistent belief.

A5 (Independent target motion). Targets move independently of each other and of the robots. They neither evade nor cooperate with the robots.

A6 (Synchronous discrete-time operation). The system operates at a fixed sampling rate $1/\tau$. The Kalman prediction step and the policy inference step are synchronized to this clock.

3.2 Robot Motion Model

Why SE(3)? Robots operating in 3-D space have both a position and an orientation. These two quantities cannot be treated independently: rotating the robot changes the direction its camera faces and therefore changes which targets fall inside its FoV. The Special Euclidean group SE(3) is the correct mathematical structure for rigid-body poses because it encodes rotation and translation jointly in a single 4×4 matrix, preserving the geometric relationships between them.

State representation. The pose of robot n at time step k is the homogeneous rigid-body transformation

$$\mathbf{x}_k^n = \begin{bmatrix} \mathbf{R}_k^n & \mathbf{p}_k^n \\ \mathbf{0}^\top & 1 \end{bmatrix} \in \text{SE}(3), \quad (3.1)$$

where $\mathbf{R}_k^n \in \text{SO}(3)$ is the 3×3 rotation matrix describing the robot’s orientation and $\mathbf{p}_k^n \in \mathbb{R}^3$ is the robot’s position, both expressed in the *global world frame*.

Kinematic update. The policy outputs a *twist* $\mathbf{a}_k^n = [\mathbf{v}_k^n; \boldsymbol{\omega}_k^n] \in \mathbb{R}^6$, which stacks the linear velocity $\mathbf{v}_k^n \in \mathbb{R}^3$ and angular velocity $\boldsymbol{\omega}_k^n \in \mathbb{R}^3$, both expressed in the robot's *body frame*. The hat map $(\cdot)^\wedge$ embeds this twist into the Lie algebra $\mathfrak{se}(3)$:

$$\hat{\mathbf{a}}_k^n = \begin{bmatrix} [\boldsymbol{\omega}_k^n]_\times & \mathbf{v}_k^n \\ \mathbf{0}^\top & 0 \end{bmatrix} \in \mathfrak{se}(3), \quad (3.2)$$

where $[\boldsymbol{\omega}_k^n]_\times$ is the 3×3 skew-symmetric matrix corresponding to $\boldsymbol{\omega}_k^n$. The discrete-time kinematic update is

$$\mathbf{x}_{k+1}^n = \mathbf{x}_k^n \exp(\tau_k \hat{\mathbf{a}}_k^n), \quad (3.3)$$

where $\tau_k = t_{k+1} - t_k$ is the sampling interval and $\exp(\cdot)$ is the matrix exponential that maps from $\mathfrak{se}(3)$ back to $\text{SE}(3)$. Right-multiplying by the incremental transform $\exp(\tau_k \hat{\mathbf{a}}_k^n)$ advances the pose by one step while keeping the result in $\text{SE}(3)$ form. This avoids the numerical drift that would occur if rotation were approximated as a flat vector updated by simple addition.

3.3 Target Motion Model

Why a linear Gaussian model? The Kalman filter is the optimal Bayesian estimator for linear Gaussian systems: it produces the exact posterior in closed form with no approximation. Any non-linearity in the target dynamics would force the use of approximate filters (EKF, UKF, particle filter), which are more complex and may not converge reliably. For the class of targets considered such as ground vehicles and pedestrians at bounded speeds, a linear near-constant-velocity model is well-established and accurate at the sampling rates used in practice.

State representation. The kinematic state of target m at time step k is $\mathbf{y}_k^m \in \mathbb{R}^d$, where the state vector stacks the target's position and velocity in the *global world frame*. For $d = 6$, one has $\mathbf{y}_k^m = [p_x, p_y, p_z, \dot{p}_x, \dot{p}_y, \dot{p}_z]^\top$ with all coordinates expressed in world-frame metres and metres per second. This global-frame representation is essential: it is the quantity that the Kalman filter estimates and that the policy receives as input. Using a world-frame state means the belief mean $\hat{\mathbf{y}}_{k|k}^m$ is directly comparable across robots and across time steps, regardless of where each robot is located.

Dynamics. The target evolves according to the linear stochastic system

$$\mathbf{y}_{k+1}^m = \mathbf{F} \mathbf{y}_k^m + \mathbf{B} \boldsymbol{\eta}_k^m + \mathbf{w}_k^m, \quad (3.4)$$

where:

- $\mathbf{F} \in \mathbb{R}^{d \times d}$ is the **state-transition matrix** that propagates the world-frame target state forward by one time step. For a near-constant velocity model with sampling interval τ , this is the block matrix $\mathbf{F} = \begin{bmatrix} \mathbf{I} & \tau \mathbf{I} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$, which adds $\tau \cdot \text{velocity}$ to the position.
- $\mathbf{B}\boldsymbol{\eta}_k^m$ captures the contribution of a known exogenous input $\boldsymbol{\eta}_k^m$ (e.g. a commanded acceleration if targets are partially controllable). In the experiments of this thesis $\boldsymbol{\eta}_k^m = \mathbf{0}$ because targets are assumed to be uncooperative.
- $\mathbf{w}_k^m \sim N(\mathbf{0}, \mathbf{Q})$ is zero-mean Gaussian **process noise** with covariance $\mathbf{Q} \in \mathbb{R}^{d \times d}$. It captures unmodeled accelerations and maneuvers. A larger $\mathbf{Q}$ tells the filter to trust predictions less and measurements more.

3.4 Sensor Model and Bayesian State Estimation

This section explains how each robot measures targets, how measurements are combined with prior predictions to maintain a belief state, and precisely which quantities are expressed in which coordinate frame at each stage of the computation.

3.4.1 Coordinate Frames

Two coordinate frames appear throughout the sensor and estimation pipeline. Confusing them is the single most common source of error in implementing multi-robot tracking systems, so they are defined precisely here.

Global world frame. A fixed, inertial reference frame shared by all robots and all targets. All quantities that are *estimated*, *stored*, or *shared* live in this frame: the true target state $\mathbf{y}_k^m$, the belief mean $\hat{\mathbf{y}}_{k|k}^m$, the belief covariance $\mathbf{P}_{k|k}^m$, the information matrix $\boldsymbol{\Lambda}_{k|k}^m$, and the robot pose $\mathbf{x}_k^n$ itself (which maps body-frame to world-frame).

Robot body frame. A frame rigidly attached to robot n , centered at its camera and aligned with the camera’s optical axis. Quantities in this frame change every time the robot moves. Only two quantities are *computed* in the body frame:

1. $\mathbf{q}_k^{m,n}$: the body-frame projection of target m ’s estimated world-frame position into robot n ’s camera, used solely to determine whether the target is inside the FoV cone.

2. The measurement $\mathbf{z}_k^m$: physically obtained in body-frame pixel coordinates by the camera, then converted to a global-frame position measurement via the known robot pose and calibration (baked into $\mathbf{H}$).

The key point is that body-frame quantities are computed at each step only to decide which targets are visible and to form the measurement residual. The filter immediately converts this information into a global-frame belief update. Everything the policy network ultimately receives, $\hat{\mathbf{y}}_{k+1|k}^m$ and $\mathbf{\Lambda}_{k+1|k}^m$, is expressed in the global world frame.

3.4.2 The Gaussian Belief State

Each robot maintains a Kalman filter that represents its knowledge about each target as a Gaussian distribution over the *global-frame* target state $\mathbf{y}_k^m$. Before incorporating the measurement at step k , the *predicted* (prior) belief is

$$p(\mathbf{y}_k^m | \mathbf{z}_{1:k-1}^m, \mathbf{x}_{0:k}) = N(\hat{\mathbf{y}}_{k|k-1}^m, \mathbf{P}_{k|k-1}^m), \quad (3.5)$$

and after incorporating the measurement at step k , the *posterior* belief is

$$p(\mathbf{y}_k^m | \mathbf{z}_{1:k}^m, \mathbf{x}_{0:k}) = N(\hat{\mathbf{y}}_{k|k}^m, \mathbf{P}_{k|k}^m). \quad (3.6)$$

In both expressions:

- $\hat{\mathbf{y}}_{k|k-1}^m \in \mathbb{R}^d$ and $\hat{\mathbf{y}}_{k|k}^m \in \mathbb{R}^d$ are the predicted and posterior **mean estimates of the target's global-frame state** (position and velocity in world coordinates). The subscript notation $k|k-1$ means “estimated at step k , having seen data up to step $k-1$ ”; $k|k$ means “estimated at step k having also incorporated the measurement at step k ”.
- $\mathbf{P}_{k|k-1}^m$ and $\mathbf{P}_{k|k}^m \in \mathbb{R}^{d \times d}$ are the predicted and posterior **covariance matrices of the estimation error**, both in global-frame units (e.g. m^2 for the position block and $(\text{m/s})^2$ for the velocity block). A larger diagonal entry means the filter is more uncertain about the corresponding state component.

3.4.3 Projecting Targets into the Robot Body Frame (FoV Check)

To determine whether target m is inside robot n 's FoV at step k , the target's estimated global-frame position must be expressed in robot n 's body frame. $\mathbf{x}_k^n \in \text{SE}(3)$ transforms from body frame to world frame; its inverse $(\mathbf{x}_k^n)^{-1}$ therefore transforms from world frame to body frame.

The projection is

$$\mathbf{q}_k^{m,n} = \mathbf{Q}_{\text{proj}}(\mathbf{x}_k^n)^{-1} \tilde{\zeta}(\mathbf{y}_k^m), \quad (3.7)$$

where:

- $\zeta(\mathbf{y}_k^m) \in \mathbb{R}^3$ extracts the **global-frame position component** from the full target state (the first three entries of $\mathbf{y}^m$).
- $\tilde{\zeta}(\mathbf{y}_k^m) = [\zeta(\mathbf{y}_k^m)^\top, 1]^\top \in \mathbb{R}^4$ is the homogeneous form required for multiplication by the 4×4 inverse pose matrix.
- $\mathbf{Q}_{\text{proj}} = [\mathbf{I}_3 \ \mathbf{0}_{3 \times 1}]$ strips the homogeneous coordinate to yield a 3-D vector.
- $\mathbf{q}_k^{m,n} \in \mathbb{R}^3$ is the resulting **body-frame position of target m as seen from robot n** . It is in body-frame units (same physical scale as the world frame, since SE(3) preserves distances). This is the quantity tested against the FoV geometry.

In practice, $\mathbf{y}_k^m$ in Eq. (3.7) is the true target position (known only in simulation). At inference time it is replaced by the world-frame estimated position $\hat{\mathbf{y}}_{k|k}^m$ to decide whether the robot should expect the target to be visible, an approximation that is accurate when estimation error is small relative to FoV size.

The set of targets currently inside at least one robot's FoV is

$$\mathcal{I}_{\mathcal{F}}(x_k, y_k) = \{ m \mid \exists n, \mathbf{q}_k^{m,n} \in \mathcal{F} \}, \quad (3.8)$$

where $\mathcal{F}$ is the triangular body-frame FoV cone defined by range r and half-angle ψ , as illustrated in Fig. 3.2.

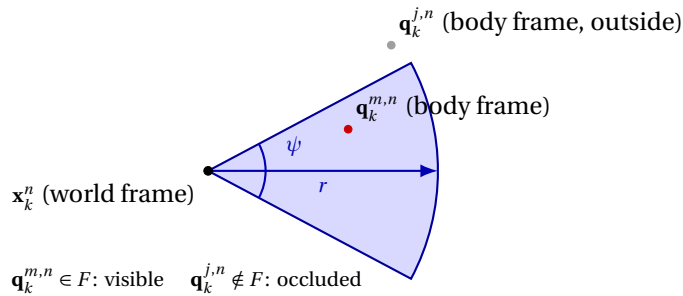


Figure 3.2: Restricted camera FoV for robot n at pose $\mathbf{x}_k^n$ (world frame). The FoV check is performed in the *robot body frame*: the global-frame target positions $\zeta(\mathbf{y}_k^m)$ and $\zeta(\mathbf{y}_k^j)$ are projected via $(\mathbf{x}_k^n)^{-1}$ to yield body-frame coordinates $\mathbf{q}_k^{m,n}$ and $\mathbf{q}_k^{j,n}$. Target m is observable (inside the FoV sector defined by r and ψ) while target j is not.

3.4.4 Measurement Model

When target $m \in \mathcal{I}_{\mathcal{F}}$ is visible, a noisy observation of its state is obtained. The measurement is modeled as a linear function of the *global-frame* target state:

$$\mathbf{z}_k^m = \mathbf{H}\mathbf{y}_k^m + \mathbf{v}_k^m, \quad \mathbf{v}_k^m \sim N(\mathbf{0}, \Sigma_v), \quad m \in I_F, \quad (3.9)$$

where:

- $\mathbf{H} \in \mathbb{R}^{p \times d}$ is the **observation matrix** that selects which components of the global-frame target state are directly observable. For a camera that measures the target's 3-D position but not its velocity, $\mathbf{H} = [\mathbf{I}_3 \ \mathbf{0}_{3 \times 3}]$, so $\mathbf{z}_k^m$ is a noisy reading of the target's global-frame position.
- $\mathbf{v}_k^m \sim N(\mathbf{0}, \Sigma_v)$ is zero-mean Gaussian **measurement noise** with covariance $\Sigma_v \in \mathbb{R}^{p \times p}$. A larger Σ_v means a less precise sensor; the filter will weight such measurements less relative to the prediction.
- $\mathbf{z}_k^m \in \mathbb{R}^p$ is the raw **measurement vector**. Although physically obtained by the camera in body-frame pixel coordinates, after applying the known robot pose $\mathbf{x}_k^n$ and the camera calibration, it is expressed as a *global-frame* quantity, specifically a noisy reading of the world-frame position $\zeta(\mathbf{y}_k^m)$. This transformation is baked into the choice of $\mathbf{H}$, so the filter operates entirely in the global frame.

No measurement when a target is occluded. When $m \notin \mathcal{I}_{\mathcal{F}}$, no measurement is available and the filter performs only a prediction step. The covariance $\mathbf{P}_{k+1|k}^m$ therefore grows due to process noise, reflecting increased uncertainty about where the target has moved. This is the fundamental coupling between robot motion, FoV coverage, and estimation uncertainty that the exploration reward exploits.

3.4.5 Kalman Filter: Predict and Update

The Kalman filter is run independently by each robot for each target $m \in \{1, \dots, T\}$. Both steps operate entirely in the **global world frame**.

Prediction (always executed at every time step). Using the target's known dynamics (Assumption A2), the filter projects the current posterior forward by one step:

$$\hat{\mathbf{y}}_{k+1|k}^m = \mathbf{F}\hat{\mathbf{y}}_{k|k}^m + \mathbf{B}\eta_k^m, \quad (3.10)$$

$$\mathbf{P}_{k+1|k}^m = \mathbf{F}\mathbf{P}_{k|k}^m\mathbf{F}^\top + \mathbf{Q}. \quad (3.11)$$

Here $\hat{\mathbf{y}}_{k+1|k}^m$ is the **predicted global-frame state**: applying $\mathbf{F}$ advances the position estimate by the expected displacement given the estimated velocity. The predicted covariance $\mathbf{P}_{k+1|k}^m$ is always at least as large as $\mathbf{P}_{k|k}^m$ because the $+\mathbf{Q}$ term adds uncertainty.

Even if the target was not visible at step k , the filter still produces a reasonable forward-looking estimate based on its last known velocity. This is why the policy receives $\hat{\mathbf{y}}_{k+1|k}^m$ (the prediction) rather than only the posterior mean: the prediction is valid even for targets currently outside every FoV.

Measurement update (executed only when target is visible). When $m \in \mathcal{J}_{\mathcal{T}}$ and measurement $\mathbf{z}_{k+1}^m$ is available, the filter corrects the prediction:

$$\mathbf{K}_{k+1}^m = \mathbf{P}_{k+1|k}^m \mathbf{H}^\top (\mathbf{H} \mathbf{P}_{k+1|k}^m \mathbf{H}^\top + \Sigma_v)^{-1}, \quad (3.12)$$

$$\hat{\mathbf{y}}_{k+1|k+1}^m = \hat{\mathbf{y}}_{k+1|k}^m + \mathbf{K}_{k+1}^m \underbrace{(\mathbf{z}_{k+1}^m - \mathbf{H} \hat{\mathbf{y}}_{k+1|k}^m)}_{\text{innovation (global frame)}}, \quad (3.13)$$

$$\mathbf{P}_{k+1|k+1}^m = (\mathbf{I} - \mathbf{K}_{k+1}^m \mathbf{H}) \mathbf{P}_{k+1|k}^m. \quad (3.14)$$

The quantities in these equations:

- $\mathbf{K}_{k+1}^m \in \mathbb{R}^{d \times p}$ is the **Kalman gain**. It is the ratio of prediction uncertainty to total uncertainty (prediction plus sensor noise). When the sensor is precise (Σ_v small), the gain is large and the filter trusts the measurement heavily. When the sensor is noisy, the gain is small and the prediction dominates.
- $\mathbf{z}_{k+1}^m - \mathbf{H} \hat{\mathbf{y}}_{k+1|k}^m$ is the **innovation**: the difference between the actual measurement and what the filter predicted it would measure. Both the measurement $\mathbf{z}_{k+1}^m$ and the predicted measurement $\mathbf{H} \hat{\mathbf{y}}_{k+1|k}^m$ are expressed in *global-frame measurement units* (e.g. metres), so the subtraction is dimensionally consistent. A large innovation means the target moved more than the dynamics predicted; the filter corrects $\hat{\mathbf{y}}$ by a fraction of this discrepancy.
- $\hat{\mathbf{y}}_{k+1|k+1}^m$ is the **posterior mean**: the best global-frame estimate of the target state after incorporating the new measurement. It is a weighted combination of the prediction and the measurement, with weights set by $\mathbf{K}$.
- $\mathbf{P}_{k+1|k+1}^m$ is the **posterior covariance**, always smaller (in the positive-semi-definite sense) than the predicted covariance: incorporating any measurement always reduces uncertainty. The more informative the measurement, the larger the reduction.

The information matrix. The inverse of the posterior covariance, $\Lambda_{k+1|k+1}^m = (\mathbf{P}_{k+1|k+1}^m)^{-1}$, is the **information matrix**. Where $\mathbf{P}$ measures uncertainty (larger means more uncertain), Λ measures certainty (larger means more certain). Its log-determinant $\log \det \Lambda$ is proportional to the negative differential entropy of the Gaussian belief, so its increase directly quantifies the bits of information gained by a measurement. This makes it the natural quantity for the exploration reward (Section 3.6.1).

Summary of coordinate frames across the pipeline. Table 3.1 makes explicit which frame each key quantity occupies at each stage.

Table 3.1: Coordinate frame of key quantities at each stage of the sensing and estimation pipeline. All estimated quantities are maintained in the global world frame. Body-frame quantities ($\mathbf{q}$, raw $\mathbf{z}$) are computed transiently for the FoV check only; $\mathbf{z}$ is converted to global-frame units before entering the filter.

Stage	Quantity	Symbol	Frame
Robot pose	Position and orientation	$\mathbf{x}_k^n$	World
Target true state	Position and velocity	$\mathbf{y}_k^m$	World
FoV check	Body-frame projection	$\mathbf{q}_k^{m,n}$	Body
Raw measurement	Pixel back-projection	(intermediate)	Body
Measurement to filter	Calibrated observation	$\mathbf{z}_k^m$	World
KF prediction	Predicted mean	$\hat{\mathbf{y}}_{k+1 k}^m$	World
KF prediction	Predicted covariance	$\mathbf{P}_{k+1 k}^m$	World
KF update	Posterior mean	$\hat{\mathbf{y}}_{k+1 k+1}^m$	World
KF update	Posterior covariance	$\mathbf{P}_{k+1 k+1}^m$	World
KF update	Information matrix	$\Lambda_{k+1 k+1}^m$	World
Policy input	Predicted mean	$\hat{\mathbf{y}}_{k+1 k}^m$	World
Policy input	Predicted info matrix	$\Lambda_{k+1 k}^m$	World

3.5 Differentiable FoV-Weighted Kalman Update

The problem with hard FoV gates. The binary indicator $\mathbb{I}(\mathbf{q}_k^{m,n} \in F)$ has zero gradient everywhere except at the FoV boundary, where the gradient is undefined. Because the Kalman measurement update is gated by this indicator, any gradient that flows back from the reward

through Λ hits a dead end at the boundary. As a result, the policy cannot learn to position robots near a target's FoV visibility edge to maximize future information gain; the reward signal provides no directional information in that critical region.

Signed distance from the FoV boundary. For any point $\mathbf{r} \in \mathbb{R}^3$ in the *robot body frame* and the body-frame FoV cone $\mathcal{F}$, define the signed distance

$$s(\mathbf{r}, F) = \begin{cases} -\min_{\mathbf{u} \in \partial F} \|\mathbf{r} - \mathbf{u}\|_2, & \mathbf{r} \in F, \\ +\min_{\mathbf{u} \in \partial F} \|\mathbf{r} - \mathbf{u}\|_2, & \mathbf{r} \notin F. \end{cases} \quad (3.15)$$

This is negative inside the FoV (how far in) and positive outside (how far out). Both $\mathbf{r}$ and $\partial \mathcal{F}$ live in the robot body frame; units are body-frame metres, normalised by σ_f below.

Soft visibility weight. The worst-case signed distance over all robots is taken for target m :

$$s_{k+1}^m = \min_n s(\mathbf{q}_{k+1}^{m,n}, F), \quad (3.16)$$

where $\mathbf{q}_{k+1}^{m,n}$ is the body-frame projection from Eq. (3.7). Taking the minimum selects the robot whose FoV comes closest to seeing target m . This scalar is passed through the complementary standard-normal CDF:

$$\alpha_{k+1}^m = 1 - \Phi\left(\frac{s_{k+1}^m}{\sigma_f}\right) \in (0, 1), \quad (3.17)$$

where $\Phi(\cdot)$ is the standard-normal CDF and $\sigma_f > 0$ controls boundary sharpness. When $s_{k+1}^m \ll 0$ (deep inside FoV), $\alpha_{k+1}^m \rightarrow 1$; when $s_{k+1}^m \gg 0$ (far outside), $\alpha_{k+1}^m \rightarrow 0$. The gradient $\partial \alpha / \partial s$ equals the Gaussian pdf, which is smooth and non-zero everywhere, enabling gradient flow through the entire pipeline.

Differentiable information update. The information-form Kalman update has a natural additive structure that accommodates the soft weight cleanly:

$$\mathbf{J}_{k+1|k+1}^m = \mathbf{J}_{k+1|k}^m + \alpha_{k+1}^m \mathbf{I}_m, \quad \mathbf{P}_{k+1|k+1}^m = \left(\mathbf{J}_{k+1|k+1}^m\right)^{-1}, \quad (3.18)$$

where:

- $\mathbf{J}_{k+1|k}^m = (\mathbf{P}_{k+1|k}^m)^{-1}$ is the **predicted information matrix** in the global frame; the starting point before incorporating the new observation.

- $\mathbf{I}_m = \mathbf{H}^\top \Sigma_v^{-1} \mathbf{H}$ is the **Fisher information contribution** of one full measurement. A noisier sensor (larger Σ_v) yields a smaller $\mathbf{I}_m$, meaning each observation adds less information.
- $\alpha_{k+1}^m \in (0, 1)$ continuously scales the information contribution. At $\alpha = 1$ (fully inside FoV) this recovers the standard Kalman update; at $\alpha = 0$ the posterior information equals the prior. The resulting $\mathbf{J}_{k+1|k+1}^m$ and $\mathbf{P}_{k+1|k+1}^m$ are global-frame quantities throughout.

This differentiable update is used *only during training*; at inference time the exact hard-FoV Kalman filter is used without approximation.

3.6 Reward Design

Each reward component is derived from a specific quantity produced by the Kalman filter. Their coordinate frames and the rationale for each mathematical form are explained alongside the equations.

3.6.1 Exploration Reward

What it measures. The exploration reward measures how much the team collectively reduced its uncertainty about all targets in a single time step. It is defined as the mean information gain across all targets:

$$r_k^{\text{exp}} = \frac{1}{T} \sum_{m=1}^T [\log \det(\Lambda_{k+1|k+1}^m) - \log \det(\Lambda_{k|k}^m)]. \quad (3.19)$$

Why this formula. The differential entropy of $\mathcal{N}(\mu, P)$ is proportional to $\log \det \mathbf{P}$. Since $\Lambda = \mathbf{P}^{-1}$, $\log \det \Lambda = -\log \det \mathbf{P}$. The difference $\log \det \Lambda_{k+1|k+1}^m - \log \det \Lambda_{k|k}^m$ is therefore the *reduction* in log-covariance-volume, equal to the reduction in differential entropy (i.e. the information gain in nats). Maximizing this over the episode is equivalent to minimizing total uncertainty, consistent with the D-optimal active-sensing criterion.

Coordinate frame. Both $\Lambda_{k+1|k+1}^m$ and $\Lambda_{k|k}^m$ are **global-frame information matrices**; the reward is a scalar and frame-independent.

Behaviour. When a robot observes a previously unseen target, the large measurement update drives a large increase in $\log \det \Lambda$, producing a high reward. When a target is outside all FoVs, only the prediction step runs: $\Lambda_{k+1|k}^m < \Lambda_{k|k}^m$ because process noise increases $\mathbf{P}$, so the reward is *negative*. This negative reward for passive steps is the key signal pushing robots to maintain target visibility.

3.6.2 Coordination Penalty

What it measures. The coordination term penalizes redundant co-observation: situations where more than one robot simultaneously observes the same target, while other targets go unobserved.

$$r_k^{\text{coord}} = \frac{1}{T \max(N-1, 1)} \sum_{m=1}^T \max\left(0, \sum_{n=1}^N \chi_k^{n,m} - 1\right), \quad (3.20)$$

where $\chi_k^{n,m} = \mathbb{I}(\mathbf{q}_k^{m,n} \in F)$ is a binary indicator equal to 1 if robot n sees target m (the **body-frame** projection $\mathbf{q}_k^{m,n}$ is inside $\mathcal{F}$) and 0 otherwise.

Why this formula. The $\max(0, \cdot)$ counts robots in excess of the first one on target m . If only one robot sees the target, the term is zero (no redundancy). Summing over all targets gives the total redundant-observation count; dividing by $T \cdot \max(N-1, 1)$ normalizes the penalty to $[0, 1]$ regardless of team size. Because it quantifies redundancy to be minimized, it is *subtracted* from the total reward in Eq. (3.32).

Coordinate frame. The FoV indicator uses the **body-frame** quantity $\mathbf{q}_k^{m,n}$. The resulting reward is a scalar.

3.6.3 Persistence Reward

What it measures. The persistence reward measures assignment consistency between adjacent time steps. A high score means robots are staying with their targets rather than switching unnecessarily.

$$r_k^{\text{pers}} = \frac{1}{T} \sum_{m=1}^T \mathbb{I}(\text{asg}_k(m) = \text{asg}_{k-1}(m)), \quad (3.21)$$

where $\text{asg}_k(m) \in \{1, \dots, N\}$ is the index of the robot assigned to target m at step k by the nearest-robot rule:

$$\text{asg}_k(m) = \arg \min_n \left\| \zeta(\hat{\mathbf{y}}_{k|k}^m) - \mathbf{p}_k^n \right\|_2. \quad (3.22)$$

Both the world-frame estimated target position $\zeta(\hat{\mathbf{y}}_{k|k}^m) \in \mathbb{R}^3$ and the world-frame robot position $\mathbf{p}_k^n \in \mathbb{R}^3$ are expressed in the **global world frame**, so their Euclidean distance has a direct physical interpretation in metres.

Why this formula. Frequent assignment switching destabilizes Kalman filter convergence: a target observed by different robots in successive steps effectively sees a different sensor geometry each time, preventing the information matrix from accumulating large values that

indicate a well-converged track. Rewarding assignment stability directly incentivizes the policy to keep each robot on its current target. The reward takes values in $[0, 1]$: 1 means all assignments are unchanged from the previous step; 0 means every assignment changed.

3.7 Baseline: Attention Policy with Linear Reward Scalarization

Policy input. Following Yang et al. [14], the policy at time step k receives

$$\mathbf{s}_k = \left[\log(\mathbf{x}_k)^\vee, \left\{ \hat{\mathbf{y}}_{k+1|k}^m \right\}_{m=1}^{T_{\max}}, \left\{ \mathbf{\Lambda}_{k+1|k}^m \right\}_{m=1}^{T_{\max}}, \mathbf{m}_k \right], \quad (3.23)$$

where each component is:

- $\log(\mathbf{x}_k)^\vee \in \mathbb{R}^6$ is the **Lie-algebra encoding of the robot's world-frame pose**. The log map projects the SE(3) pose matrix to a 6-vector in $\mathfrak{se}(3)$, and $(\cdot)^\vee$ extracts its coordinate vector. This provides a compact, numerically stable encoding of the robot's position and orientation in the world frame without the redundancy of a 4×4 matrix.
- $\hat{\mathbf{y}}_{k+1|k}^m \in \mathbb{R}^d$ is the **one-step-ahead Kalman prediction of target m 's global-frame state**. Using the prediction rather than the posterior gives the policy a forward-looking view: even if the target was just observed, the policy reasons about where it will be next step, which is what matters for generating the current velocity command.
- $\mathbf{\Lambda}_{k+1|k}^m \in \mathbb{R}^{d \times d}$ is the **predicted information matrix for target m** in the global frame. High $\mathbf{\Lambda}$ (small $\mathbf{P}$) means a well-tracked target; low $\mathbf{\Lambda}$ (large $\mathbf{P}$) means the target urgently needs re-observation.
- $\mathbf{m}_k \in \{0, 1\}^{T_{\max}}$ is the **padding mask**: $m_k^j = 1$ for $j \leq T$ (active) and $m_k^j = 0$ for $j > T$ (padded). This allows the architecture to be fixed at $T_{\max}$ slots while operating on any number of actual targets up to $T_{\max}$ without retraining.

Network architecture.

$$\mathbf{e}_k^a = f_a(\log(\mathbf{x}_k)^\vee), \quad (3.24)$$

$$\mathbf{E}_k^t = f_t\left(\{\hat{\mathbf{y}}_{k+1|k}^m, \mathbf{\Lambda}_{k+1|k}^m\}_{m=1}^{T_{\max}}\right), \quad (3.25)$$

$$\mathbf{S}_k = \frac{\mathbf{e}_k^a (\mathbf{E}_k^t)^\top}{\beta_{\text{att}}}, \quad (3.26)$$

$$\tilde{\mathbf{S}}_k = (1 + \log \mathbf{m}_k) \odot \mathbf{S}_k, \quad (3.27)$$

$$\mathbf{A}_k = \text{softmax}(\tilde{\mathbf{S}}_k), \quad (3.28)$$

$$\mathbf{h}_k = \text{ReLU}(f_{\text{fuse}}([\mathbf{e}_k^a, \mathbf{A}_k \mathbf{E}_k^t])), \quad (3.29)$$

$$\mathbf{a}_k = f_o(\mathbf{h}_k), \quad (3.30)$$

$$\mathbf{u}_k = f_{\text{scale}}(\mathbf{a}_k). \quad (3.31)$$

The meaning of each step:

- **Eq. (3.24):** The **robot encoder** f_a (two-layer ReLU MLP) compresses the 6-D world-frame pose encoding into a fixed-size embedding $\mathbf{e}_k^a$, the “query” in the subsequent attention.
- **Eq. (3.25):** The **target encoder** f_t maps each target’s concatenated world-frame features $[\hat{\mathbf{y}}_{k+1|k}^m; \mathbf{\Lambda}_{k+1|k}^m]$ to an embedding. The rows of $\mathbf{E}_k^t$ form the “key-value” matrix.
- **Eq. (3.26):** The **attention score** $\mathbf{S}_k$ is the dot product of the robot embedding with each target embedding, scaled by temperature β_{att} to prevent softmax saturation.
- **Eq. (3.27):** The **mask fill** sets padded target slots to $-\infty$ via $\log(0) = -\infty$, ensuring padded slots receive zero attention weight.
- **Eq. (3.28):** The **attention weights** $\mathbf{A}_k$ form a normalised distribution over T active targets, representing how much the policy focuses on each target.
- **Eq. (3.29):** The **fusion layer** f_{fuse} concatenates the robot embedding with the attention-weighted target context $\mathbf{A}_k \mathbf{E}_k^t$ and passes through a ReLU MLP to produce the shared latent $\mathbf{h}_k$. This is the most important intermediate representation: it is re-used by the DRA’s critic heads.
- **Eqs. (3.30)–(3.31):** The **output head** f_o maps $\mathbf{h}_k$ to a raw action via a two-layer Tanh MLP. The scaling function f_{scale} maps the Tanh output to physically meaningful velocity ranges, producing $\mathbf{u}_k = [\mathbf{v}_k^\top, \boldsymbol{\omega}_k^\top]^\top$ in body-frame velocity units.

The full data flow with block identifiers B1–B11 is shown in Fig. 3.3 and described in Appendix B.

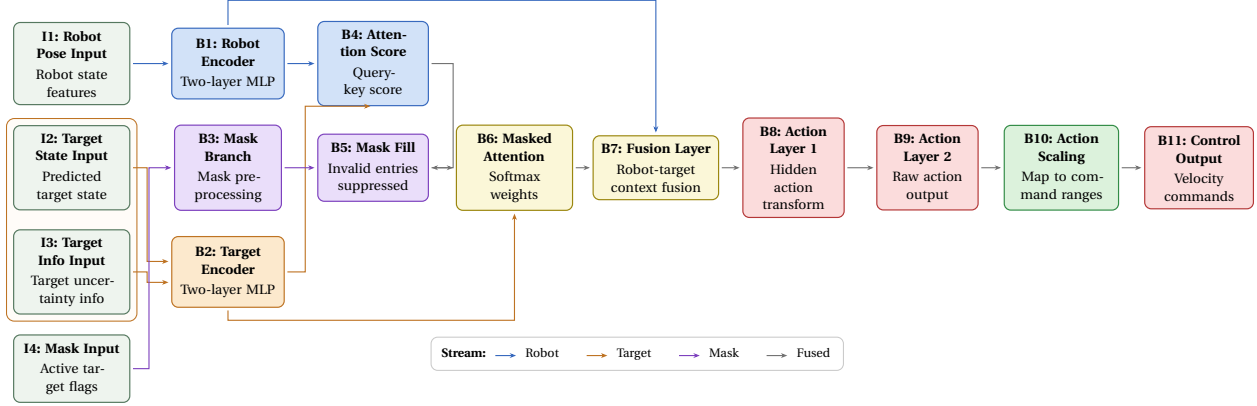


Figure 3.3: Baseline policy network with block identifiers (B1–B11) and a shared masked-attention backbone under linear reward scalarization.

Linear reward scalarization. The three reward terms are combined into a single training objective:

$$R_{\text{total}} = \alpha_{\text{exp}} \bar{r}^{\text{exp}} + \alpha_{\text{pers}} \bar{r}^{\text{pers}} - \alpha_{\text{coord}} \bar{r}^{\text{coord}}, \quad (3.32)$$

where $\bar{r}^{(\cdot)} = \frac{1}{K} \sum_{k=1}^K r_k^{(\cdot)}$ is the episode mean of each component and $\alpha_{(\cdot)} \geq 0$ are non-negative scalar weights. The coordination term is subtracted because it quantifies redundancy the policy should minimize. The policy is trained by gradient ascent on R_{total} , equivalently minimizing $\mathcal{L}_{\text{base}} = -R_{\text{total}}$.

3.8 Decomposed Reward Architecture

Why linear scalarization fails. Linear scalarization is simple, but it is the main source of training instability when reward terms compete. The exploration reward encourages robots to leave currently tracked targets to reduce long-horizon uncertainty; the persistence reward rewards keeping assignments unchanged. These two objectives produce gradients that point in opposite directions at the same weight updates, and their relative magnitudes shift as training progresses. The result is a scalarized objective sensitive to the choice of $\alpha_{(\cdot)}$ coefficients and prone to producing oscillatory policies that never settle on stable tracking behavior.

The decomposed critic approach. To resolve this, the baseline is extended from pure policy gradient to a deterministic actor-critic framework following the Hybrid Reward Architecture of van Seijen et al. [37]. Rather than a single value function trained on R_{total} , a separate Q-head is attached to the shared encoder g_ω for each reward component. A shared deterministic policy head learns to satisfy all three simultaneously.

3.8.1 Decomposed Critics

For each component $j \in \{\text{exp}, \text{coord}, \text{pers}\}$, a dedicated Q-head $Q_{\theta_j}^{(j)}$ takes the shared latent $\mathbf{h}_k$ and proposed action $\mathbf{u}$, and estimates the expected discounted return for that component alone:

$$Q_{\theta_j}^{(j)}(\mathbf{h}_k, \mathbf{u}) \approx \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_{k+t}^{(j)} \mid \mathbf{h}_k, \mathbf{u}_k = \mathbf{u} \right], \quad (3.33)$$

where $\gamma \in (0, 1)$ is the discount factor that controls the trade-off between immediate and long-term reward. Each Q-head is a two-layer MLP taking $[\mathbf{h}_k; \mathbf{u}]$ as input and outputting a scalar.

TD learning per head. Each head is updated using its own one-step Temporal Difference (TD) target:

$$y_k^{(j)} = r_k^{(j)} + \gamma Q_{\bar{\theta}_j}^{(j)}(\mathbf{h}_{k+1}, \pi_\phi(\mathbf{h}_{k+1})), \quad (3.34)$$

where $Q_{\bar{\theta}_j}^{(j)}$ is a periodically synchronized *target network* (a slowly updated copy with parameters $\bar{\theta}_j$) used to stabilize training. The critic loss is

$$\mathcal{L}_{\text{critic}} = \sum_j \mathbb{E} \left[\left(y_k^{(j)} - Q_{\theta_j}^{(j)}(\mathbf{h}_k, \mathbf{u}_k) \right)^2 \right]. \quad (3.35)$$

Because each head receives only its own reward signal $r_k^{(j)}$, the exploration head cannot be overwhelmed by the persistence gradient and vice versa.

3.8.2 Policy Head

A deterministic policy head π_ϕ maps $\mathbf{h}_k$ to a control command in one forward pass:

$$\mathbf{u}_k = \pi_\phi(\mathbf{h}_k). \quad (3.36)$$

The aggregate action-value across all decomposed heads is

$$Q_k^{\text{DRA}}(\mathbf{u}) = \sum_j Q_{\theta_j}^{(j)}(\mathbf{h}_k, \mathbf{u}), \quad (3.37)$$

and the policy head is trained by gradient ascent through the frozen Q-heads:

$$\mathcal{L}_{\text{actor}} = -\mathbb{E}[Q_k^{\text{DRA}}(\pi_\phi(h_k))]. \quad (3.38)$$

The gradient $\partial \mathcal{L}_{\text{actor}} / \partial \phi$ passes through Q_k^{DRA} and back into π_ϕ , updating the policy to produce actions that all three Q-heads rate highly. Because the Q-heads are frozen during the actor update, each head can only push the policy toward its own objective without suppressing the others.

3.8.3 Two-Stage Training Procedure

Training alternates between two stages at each iteration:

1. **Stage 1 — Critic update.** Minimize $\mathcal{L}_{\text{critic}}$ by updating θ_j for all j . The shared encoder g_ω and policy head π_ϕ are held fixed. Each Q-head independently improves its estimate of the component return.
2. **Stage 2 — Actor update.** Minimize $\mathcal{L}_{\text{actor}}$ by updating ϕ (policy head) and ω (shared encoder). The Q-heads θ_j are frozen. Gradient flows from Q_k^{DRA} back through π_ϕ and g_ω , adjusting the latent representation to support all three objectives simultaneously.

This procedure is directly analogous to DDPG [42], extended to a decomposed multi-head critic. Figure 3.4 shows the complete data flow with Stage 1 and Stage 2 regions annotated; block identifiers D1–D13 are described in Appendix B.

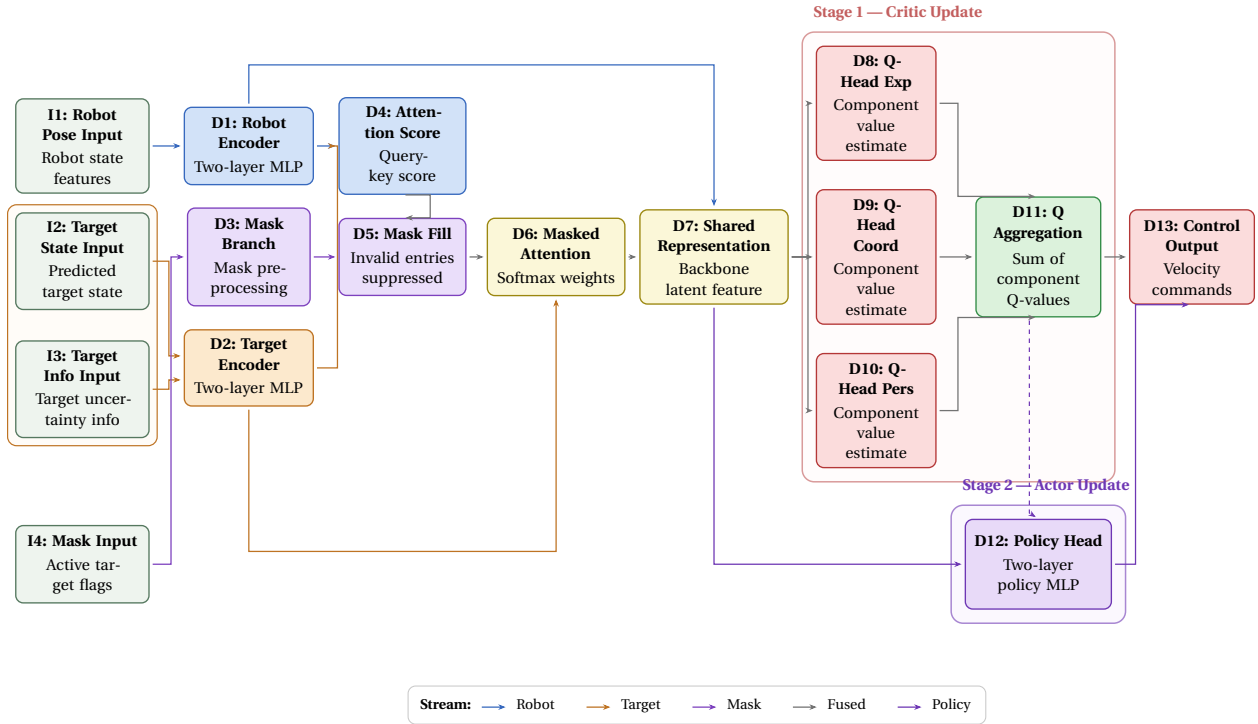


Figure 3.4: Decomposed Reward Architecture (DRA) with block identifiers D1–D13. The shared backbone (D1–D7) is identical to the baseline attention policy. Three independent Q-heads (D8–D10) are trained in Stage 1 (Critic Update) using per-component TD targets. The aggregation block (D11) sums the Q-head outputs. The dedicated policy head (D12) is updated in Stage 2 (Actor Update) by back-propagating through the frozen Q-heads, ensuring each objective contributes independently to the policy gradient.

CHAPTER

4

EXPERIMENTS AND RESULTS

This chapter presents the experimental evaluation of the proposed Decomposed Reward Architecture (DRA). Section 4.1 describes the hardware and evaluation pipeline. Section 4.2 presents qualitative results for successful DRA tracking sequences and the Gazebo simulation. Section 4.3 presents quantitative comparisons across all methods. Section 4.4 analyzes the overall results. Section 4.5 examines failure cases of the DRA policy in detail. The qualitative comparison of the linear reward baseline is provided in Appendix C for reference.

4.1 Experimental Setup

4.1.1 Hardware and Software Stack

All experiments were conducted on a laptop equipped with an Intel Core i9 CPU and an NVIDIA GeForce RTX 4090 GPU (Alienware m18 R2). Evaluation followed a two-stage pipeline: (1) quantitative testing in a synthetic simulator with `matplotlib`-based rendering, and (2) policy verification in Gazebo + ROS 2 using the same checkpoints and scenario setup.

4.1.2 Evaluation Protocol

Three methods are evaluated over 30 test episodes each:

1. **Yang et al. [14]**: the attention policy baseline with a single scalar reward.
2. **Linear reward combination (ours)**: the same attention policy architecture trained with the linearly scalarized reward R_{total} from Eq. (3.32).
3. **Decomposed Reward Architecture (DRA, ours)**: the proposed actor-critic framework with dedicated critic heads.

For qualitative diagnosis, each sequence is represented by a horizontal strip of four frames sampled at equal intervals, and multiple sequences are stacked vertically for comparison. This format is used consistently across all qualitative figures.

4.1.3 Metrics

Metrics are computed from per-episode field-of-view observations and aggregated across episodes:

- **Cumulative tracking (%)**: fraction of time steps during which each target is observed by at least one robot, averaged over all targets.
- **Overlap (%)**: fraction of time steps during which a target is simultaneously observed by more than one robot, averaged over all targets. High overlap indicates redundant sensing effort.
- **Exclusive tracking (%)**: cumulative tracking minus overlap; captures effective non-redundant coverage.
- **Mean unique targets tracked**: mean number of distinct targets observed by at least one robot per time step.

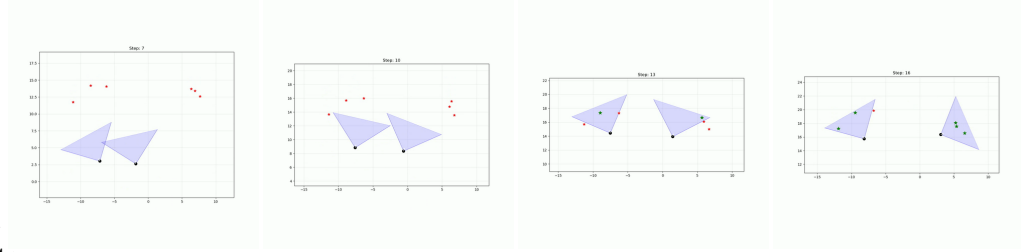
4.2 Qualitative Results

4.2.1 Synthetic Simulator Results

Figure 4.1 presents two successful DRA sequences: Sequence S1 (6 targets) and Sequence S2 (5 targets). Each row shows four frames sampled at equal intervals of the episode stacked horizontally. Both sequences exhibit sustained target commitment once tracks are established,

with distinct spatial assignments maintained throughout and minimal unnecessary handoffs between robots.

Sequence S1
DRA, 6 targets



Sequence S2
DRA, 5 targets

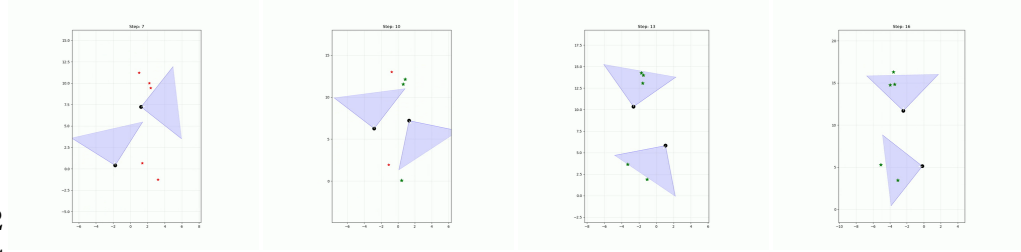


Figure 4.1: Qualitative successful tracking sequences for the DRA policy. Each row corresponds to one sequence and shows four frames sampled from equal intervals stacked horizontally. Robots maintain distinct target assignments with minimal handoffs in both sequences.

4.2.2 Gazebo Simulation Results

Figure 4.2 reports a representative rollout from the Gazebo + ROS 2 simulation environment using the same policy checkpoint and scenario setup as the synthetic evaluations. This confirms that the policy transfers directly from the `matplotlib` training environment to a physics-based simulator without modification.

Sequence GZ1

Gazebo
simulation



Figure 4.2: Qualitative results from Gazebo simulation. Sequence GZ1 shows three consecutive frames from a representative rollout under the same policy and scenario setup used in synthetic evaluation.

4.3 Quantitative Results

4.3.1 Overall Performance

Table 4.1 reports episode-level mean $\pm$ standard deviation across all 30 test episodes per method. The DRA policy achieves the highest cumulative tracking rate (57.71%) and the largest mean number of tracked targets (2.1821), improving cumulative tracking by 45.9% over the linear reward combination policy and by 154.9% over Yang et al. [14].

Table 4.1: Overall quantitative comparison across 30 test episodes per method. Values are reported as mean $\pm$ standard deviation. Bold denotes the best result per metric.

Metric	Yang et al. [14]	Linear reward combination (ours)	DRA (ours)
Cumulative tracking (%)	22.64 ± 8.44	39.56 ± 13.29	57.71 ± 20.89
Overlap (%)	0.30 ± 0.73	2.70 ± 4.69	21.98 ± 28.64
Exclusive tracking (%)	22.34 ± 8.50	36.87 ± 13.01	35.73 ± 15.87
Mean unique targets tracked	0.7271 ± 0.3177	1.7388 ± 0.7248	2.1821 ± 0.7454

This improvement is accompanied by higher overlap (21.98%), indicating more frequent redundant co-observation. Consequently, the exclusive tracking metric remains close to the linear model, indicating that the additional cumulative coverage is partially offset by cooperative re-observation. The elevated overlap is a clear target for improvement in future policy

refinement.

4.3.2 Performance by Target Count

Table 4.2 breaks down cumulative tracking by the realized number of targets. The DRA policy consistently attains the best performance for 4, 5, and 6 targets, with the largest absolute margin in the 6-target setting (66.84% vs. 29.60% for the linear policy). For 7 targets, DRA and the linear reward combination perform similarly, suggesting that the coordination head alone may not be sufficient to suppress redundant co-observation in high-density configurations.

Table 4.2: Cumulative tracking percentage (%) grouped by realized target count. Values are reported as mean $\pm$ standard deviation.

Method	4 targets	5 targets	6 targets	7 targets
Yang et al. [14]	28.93 $\pm$ 10.03	25.02 $\pm$ 5.91	16.82 $\pm$ 1.17	16.80 $\pm$ 3.76
Linear reward combination (ours)	43.05 $\pm$ 14.11	44.27 $\pm$ 8.14	29.60 $\pm$ 4.06	38.24 $\pm$ 16.84
DRA (ours)	66.79 $\pm$ 22.86	58.71 $\pm$ 11.72	66.84 $\pm$ 22.39	38.23 $\pm$ 11.09

4.4 Result Analysis

The quantitative comparisons confirm that the DRA policy provides the strongest overall tracking performance among the evaluated methods. The DRA attains the highest cumulative tracking rate (57.71%) and the largest mean number of unique targets tracked (2.1821), versus 39.56% and 1.7388 for the linear reward combination, and 22.64% and 0.7271 for Yang et al. [14]. These results support the central claim that decomposing competing reward objectives into separate critic heads yields better optimization outcomes than direct linear scalarization.

The successful qualitative sequences in Fig. 4.1 demonstrate all three intended behaviors: *persistence* (sustained target dwell times once a track is established), *coordination* (distinct spatial assignments with minimal unnecessary handoffs), and *exploration* (continued target acquisition across the full episode duration). The Gazebo results in Fig. 4.2 further confirm that these behaviors transfer to a physics-based simulator without additional fine-tuning.

Across target-count breakdowns (Table 4.2), the DRA advantage is clearest in the 4–6 target regime.

Table 4.3: Overlap percentage (%) for the DRA policy, broken down by target count. Values are reported as mean $\pm$ standard deviation.

Setting	Overlap (%)
Overall (all episodes)	21.98 $\pm$ 28.64
4 targets	40.55 $\pm$ 33.41
5 targets	11.29 $\pm$ 7.54
6 targets	35.00 $\pm$ 35.96
7 targets	4.15 $\pm$ 7.36

4.5 Failure Case Analysis

While the DRA policy achieves strong overall performance, it is not without failure modes. This section identifies the primary behavioral failure patterns, presents representative qualitative sequences, and discusses the root causes and directions for future mitigation. The qualitative failure behavior of the linear reward baseline, which exhibits a distinct and complementary failure mode, is documented separately in Appendix C.

4.5.1 Observed Failure Modes

Two failure modes recur consistently across DRA evaluation runs:

Cluster convergence. In moderate-density target configurations (4–6 targets), robots initially spread across the environment but gradually converge toward the largest spatial cluster of targets. Once a cluster is established, the per-step information gain from that cluster is large because multiple targets simultaneously contribute to $\Delta \log \det \mathbf{\Lambda}$, making it locally rational for additional robots to join. The result is high overlap within the cluster and zero coverage of outlier targets.

Coverage collapse at high target density. At 7 targets, the DRA cumulative tracking rate (38.23%) drops sharply and approaches the linear baseline. Robots spread out initially but fail to maintain simultaneous coverage as target positions evolve. This suggests limited generalization of the coordination head to the highest-density regime encountered during evaluation.

4.5.2 Overlap Analysis for DRA

Table 4.3 reports overlap statistics for the DRA policy. The highest overlap occurs in the 4-target (40.55%) and 6-target (35.00%) regimes. The relatively low overlap in the 7-target setting (4.15%)

is consistent with weaker cumulative tracking at that density, suggesting that robots spread out but fail to maintain simultaneous coverage.

4.5.3 Qualitative Failure Sequences

Figure 4.3 presents two representative DRA failure sequences, F1 and F2. In Sequence F1, robots spread across the environment in the early frames but converge toward the densest cluster by the third frame, leaving outlier targets unobserved for the remainder of the episode. In Sequence F2, cluster collapse occurs earlier and is more severe, with most robots concentrated on a two-target subgroup by the second frame.

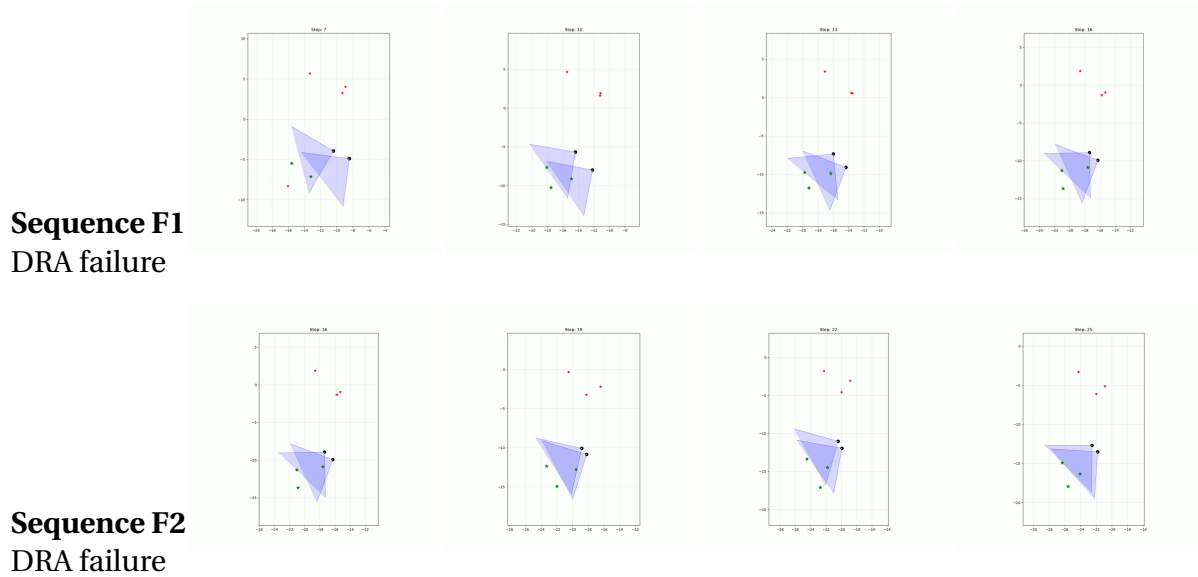


Figure 4.3: DRA failure sequences F1 and F2. In both cases, robots converge toward the nearest target cluster, reducing team-level coverage and increasing redundant co-observation. This cluster-convergence pattern is the primary remaining limitation of the current DRA policy and is consistent with the elevated overlap values reported in Table 4.3.

4.5.4 Discussion

The cluster-convergence failure reveals a structural gap in the current reward design: the coordination penalty operates at the level of individual target–robot pairs and does not carry a global signal about which targets in the full set are currently uncovered. A natural direction for future work is to augment the coordination penalty with a global coverage term that explicitly

rewards uniform target set coverage, or to incorporate a counterfactual credit assignment mechanism [38] that attributes the information gain from a cluster observation only to the robot that would have been most useful there, redirecting remaining robots toward uncovered targets.

CHAPTER

5

CONCLUSION AND FUTURE WORK

5.1 Summary of Results

This thesis presented a hybrid framework for collaborative active tracking that combines Bayesian state-estimation structure with a Decomposed Reward Architecture (DRA). The proposed formulation jointly addresses three objectives: uncertainty-aware exploration, inter-robot coordination, and tracking persistence. By assigning competing reward components to dedicated critic heads within an actor-critic framework, the method improves optimization stability while preserving computationally efficient policy inference at runtime.

Experimental results in a synthetic multi-robot tracking simulator demonstrate that the DRA policy achieves a cumulative tracking rate of 57.71%, representing a 45.9% improvement over the linear reward combination baseline and a 154.9% improvement over the Yang et al. [14] attention policy. The DRA policy also achieves the largest mean number of unique targets tracked (2.1821 per step) across all evaluated methods. Policy transfer to a Gazebo + ROS 2 simulation environment confirmed that the learned behaviors generalize to a physics-based simulator without additional fine-tuning.

5.2 Overarching Conclusions

Three overarching conclusions follow from this work.

Reward decomposition improves tracking stability. The primary limitation of linear reward scalarization is that conflicting gradient signals from exploration and persistence objectives destabilize training, producing oscillatory policies that frequently switch target assignments. The DRA resolves this by isolating each reward component to its own critic head, allowing each objective to be optimized without interference from competing gradients. The qualitative evidence, shorter dwell intervals and more frequent handoffs under the linear baseline, directly corroborates this conclusion.

Bayesian estimation provides a principled reward signal. Grounding the exploration reward in information gain from the Kalman filter information matrix gives the policy a signal that is directly tied to localization uncertainty reduction. This is in contrast to heuristic exploration bonuses that do not distinguish between observing a well-tracked target and observing one with high uncertainty.

Overlap remains an open challenge. While the DRA achieves the highest cumulative tracking rate, it also incurs the highest overlap (21.98%), particularly in 4-target and 6-target configurations. This indicates that the coordination critic head alone is insufficient to fully suppress redundant co-observation. Future work should focus specifically on strengthening the coordination signal during training.

5.3 Future Work

Future work will focus on two immediate directions.

First, the framework will be deployed on physical multi-robot platforms to validate robustness under sensing noise, communication constraints, calibration drift, and dynamic environmental occlusions. The Gazebo transfer results are encouraging, but real-world deployment introduces sensor artifacts and latency that are not modeled in simulation.

Second, the experimental evaluation will be expanded through broader ablation analysis and increased statistical coverage across scenarios. Specific ablations of interest include: (i) ablating individual reward components to quantify their independent contributions; (ii) varying the number of robots N to assess scalability; and (iii) evaluating alternative decomposition strategies, such as per-target critic heads rather than per-objective heads.

CHAPTER

6

ACRONYMS

A summary of all acronyms used in this thesis is provided in Table 6.1.

Table 6.1: Acronyms used in this thesis, listed in alphabetical order.

Acronym	Expansion
CDF	Cumulative Distribution Function
DRA	Decomposed Reward Architecture
FoV	Field of View
GNN	Graph Neural Network
IMU	Inertial Measurement Unit
KF	Kalman Filter
LLM	Large Language Model
MRS	Multi-Robot System
NP	Nondeterministic Polynomial (as in NP-hard)
OSPA	Optimal Sub-Pattern Assignment
PHD	Probability Hypothesis Density
RL	Reinforcement Learning
ROS	Robot Operating System
SE(3)	Special Euclidean Group in Three Dimensions

Table 6.1 (continued)

Acronym	Expansion
SO(3)	Special Orthogonal Group in Three Dimensions
TD	Temporal Difference
UGV	Unmanned Ground Vehicle

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APPENDICES

APPENDIX

A

NOTATION AND VARIABLES

A summary of the mathematical notation and variables used throughout this thesis is provided in Table A.1.

Table A.1: Mathematical notation and variables used in this thesis, listed by category.

Symbol	Description
<i>Indices and Scalars</i>	
N	Number of robots in the team
T	Number of moving targets
$T_{\max}$	Maximum number of targets (for padding)
K	Episode length (number of time steps)
k	Discrete time step index
n	Robot index, $n \in \{1, \dots, N\}$
m	Target index, $m \in \{1, \dots, T\}$
d	Dimension of target state space
r	Camera sensing range
ψ	Camera FoV half-angle
γ	Discount factor
τ_k	Sampling interval $t_{k+1} - t_k$

Table A.1 (continued)

Symbol	Description
σ_f	FoV boundary sharpness parameter
<i>Robot State</i>	
$\mathbf{x}_k^n \in \text{SE}(3)$	Pose (position and orientation) of robot n at step k
$\mathbf{R}_k^n \in \text{SO}(3)$	Rotation matrix component of robot n 's pose
$\mathbf{p}_k^n \in \mathbb{R}^3$	Position component of robot n 's pose
$\mathbf{a}_k^n \in \mathbb{R}^6$	Twist (linear and angular velocities) of robot n
$\mathbf{v}_k^n \in \mathbb{R}^3$	Linear velocity of robot n
$\boldsymbol{\omega}_k^n \in \mathbb{R}^3$	Angular velocity of robot n
$\mathbf{u}_k$	Control output (velocity command)
<i>Target State and Estimation</i>	
$\mathbf{y}_k^m \in \mathbb{R}^d$	State of target m at step k
$\mathbf{F} \in \mathbb{R}^{d \times d}$	State-transition matrix
$\mathbf{B}$	Control input mapping matrix
$\boldsymbol{\eta}_k^m$	Exogenous control input for target m
$\mathbf{w}_k^m$	Process noise, $\mathbf{w}_k^m \sim N(\mathbf{0}, \mathbf{Q})$
$\mathbf{Q}$	Process noise covariance
$\mathbf{z}_k^m$	Measurement of target m at step k
$\mathbf{H}$	Observation model matrix
$\mathbf{v}_k^m$	Measurement noise, $\mathbf{v}_k^m \sim N(\mathbf{0}, \boldsymbol{\Sigma}_v)$
$\boldsymbol{\Sigma}_v$	Measurement noise covariance
$\hat{\mathbf{y}}_{k k}^m$	Posterior mean estimate of target m at step k
$\mathbf{P}_{k k}^m$	Posterior covariance of target m at step k
$\hat{\mathbf{y}}_{k+1 k}^m$	Predicted (prior) mean of target m
$\mathbf{P}_{k+1 k}^m$	Predicted (prior) covariance of target m
$\mathbf{K}_{k+1}^m$	Kalman gain for target m at step $k + 1$
$\sim_{k k}^m$	Information matrix ($= \mathbf{P}_{k k}^{m,-1}$)
$\mathbf{J}_{k k}^m$	Information matrix used in soft FoV update
$\mathbf{I}_m$	Fisher information contribution $\mathbf{H}^\top \boldsymbol{\Sigma}_v^{-1} \mathbf{H}$
α_{k+1}^m	Soft FoV visibility weight for target m
<i>Policy and Reward</i>	
$\mathcal{F}$	Camera FoV set
$\mathcal{I}_{\mathcal{F}}$	Set of visible target indices
$\mathbf{q}_k^{m,n}$	Position of target m in robot n 's body frame
$\chi_k^{n,m}$	Indicator: 1 if robot n observes target m , else 0

Table A.1 (continued)

Symbol	Description
$\text{asg}_k(m)$	Robot assignment of target m at step k
r_k^{exp}	Per-step exploration reward
r_k^{coord}	Per-step coordination penalty
r_k^{pers}	Per-step persistence reward
R_{total}	Total linearly scalarized reward
$\alpha_{\text{exp}}, \alpha_{\text{pers}}, \alpha_{\text{coord}}$	Reward weighting coefficients
<i>Network Architecture</i>	
$\mathbf{s}_k$	Full policy input state at step k
$\mathbf{m}_k$	Padding mask vector
$\mathbf{e}_k^a$	Robot encoder output embedding
$\mathbf{E}_k^t$	Target encoder output embeddings
$\mathbf{h}_k$	Shared encoder output (latent representation)
$\mathcal{G}_\omega$	Shared encoder mapping $\mathbf{s}_k \mapsto \mathbf{h}_k$
π_ϕ	Policy head mapping $\mathbf{h}_k \mapsto \mathbf{u}_k$
$Q_{\theta_j}^{(j)}$	Q-head for reward component j
Q_k^{DRA}	Aggregated DRA action-value
β_{att}	Attention temperature
$\mathcal{L}_{\text{base}}$	Baseline policy gradient loss
$\mathcal{L}_{\text{critic}}$	DRA critic loss
$\mathcal{L}_{\text{actor}}$	DRA actor loss

APPENDIX

B

ARCHITECTURE BLOCK DIAGRAM DESCRIPTIONS

This appendix provides detailed descriptions of the block identifiers used in the architecture diagrams for the baseline policy network and the Decomposed Reward Architecture (DRA). These descriptions are intended as a companion to the diagrams referenced in Chapter 3.

B.1 Baseline Policy Architecture (Blocks B1–B11)

The baseline policy network processes three input streams—robot pose, target state and uncertainty, and a padding mask—through a shared masked-attention backbone and outputs a continuous control command. Table B.1 describes each block.

Table B.1: Block descriptions for the baseline policy network (B1–B11).

Block	Name	Description
I1	Robot Pose Input	Robot state features derived from $\log(\mathbf{x}_k)^\vee$.
I2	Target State Input	Predicted target mean $\hat{\mathbf{y}}_{k+1 k}^m$ for each slot.

Table B.1 (continued)

Block	Name	Description
I3	Target Info Input	Predicted information matrix $\tilde{\mathbf{m}}_{k+1 k}^m$ for each slot.
I4	Mask Input	Binary padding mask $\mathbf{m}_k$ indicating active target slots.
B1	Robot Encoder	Two-layer MLP with ReLU activations mapping robot pose features to embedding $\mathbf{e}_k^a$. Equation (3.24).
B2	Target Encoder	Two-layer MLP mapping concatenated per-target features $[\hat{\mathbf{y}}_{k+1 k}^m; \tilde{\mathbf{m}}_{k+1 k}^m]$ to embeddings $\mathbf{E}_k^t$. Equation (3.25).
B3	Mask Branch	Preprocessing of the padding mask for use in B5.
B4	Attention Score	Query-key dot-product score $\mathbf{S}_k = \mathbf{e}_k^a(\mathbf{E}_k^t)^\top / \beta_{\text{att}}$. Equation (3.26).
B5	Mask Fill	Suppresses invalid (padded) target slots by applying the mask: $\tilde{\mathbf{S}}_k = (1 + \log \mathbf{m}_k) \odot \mathbf{S}_k$. Equation (3.27).
B6	Masked Attention	Softmax over masked scores to produce attention weights $\mathbf{A}_k$. Equation (3.28).
B7	Fusion Layer	Concatenates robot embedding and attention-weighted target context; applies ReLU MLP to produce shared latent $\mathbf{h}_k$. Equation (3.29).
B8	Action Layer 1	First hidden layer of the action MLP f_o ; applies Tanh activation.
B9	Action Layer 2	Second hidden layer of the action MLP f_o ; produces raw action output $\mathbf{a}_k$. Equation (3.30).
B10	Action Scaling	Maps raw action $\mathbf{a}_k$ to valid velocity command ranges via f_{scale} .
B11	Control Output	Final velocity command $\mathbf{u}_k = [(\mathbf{v}_k)^\top, (\boldsymbol{\omega}_k)^\top]^\top$. Equation (3.31).

B.2 Decomposed Reward Architecture (Blocks D1–D13)

The DRA reuses the shared backbone from the baseline (D1–D7, functionally identical to B1–B7) and adds three reward-specific Q-heads, an aggregation block, and a dedicated policy head. Table B.2 describes each block.

Table B.2: Block descriptions for the Decomposed Reward Architecture (D1–D13).

Block	Name	Description
D1	Robot Encoder	Same as B1. Maps robot pose features to $\mathbf{e}_k^a$.
D2	Target Encoder	Same as B2. Maps per-target features to $\mathbf{E}_k^t$.
D3	Mask Branch	Same as B3. Preprocesses the padding mask.
D4	Attention Score	Same as B4. Computes query-key scores.
D5	Mask Fill	Same as B5. Suppresses padded target slots.
D6	Masked Attention	Same as B6. Produces attention weights $\mathbf{A}_k$.
D7	Shared Representation	Same as B7. Produces shared latent $\mathbf{h}_k = g_\omega(\mathbf{s}_k)$.
D8	Q-Head (Exploration)	Two-layer MLP critic head estimating expected discounted return for r^{exp} . Equation (3.33). Updated in Stage 1.
D9	Q-Head (Coordination)	Two-layer MLP critic head estimating expected discounted return for r^{coord} . Equation (3.33). Updated in Stage 1.
D10	Q-Head (Persistence)	Two-layer MLP critic head estimating expected discounted return for r^{pers} . Equation (3.33). Updated in Stage 1.
D11	Q Aggregation	Sums the three component Q-values to produce $Q_k^{\text{DRA}}(\mathbf{u})$. Equation (3.37). Updated in Stage 1.

Table B.2 (continued)

Block	Name	Description
D12	Policy Head	Two-layer MLP with Tanh activations mapping $\mathbf{h}_k$ to control command $\mathbf{u}_k = \pi_\phi(\mathbf{h}_k)$. Equation (3.36). Updated in Stage 2.
D13	Control Output	Final velocity command output from D12.

Training stage summary. During Stage 1 (Critic Update), blocks D8, D9, D10, and D11 are updated using the TD targets from Equation (3.34) and the critic loss $\mathcal{L}_{\text{critic}}$ from Equation (3.35); blocks D1–D7 and D12 are held fixed. During Stage 2 (Actor Update), block D12 and the shared encoder D1–D7 are updated using the actor loss $\mathcal{L}_{\text{actor}}$ from Equation (3.38); blocks D8–D11 are held fixed (frozen Q-heads).

APPENDIX

C

LINEAR REWARD BASELINE: QUALITATIVE FAILURE ANALYSIS

This appendix documents the qualitative failure behavior of the linear reward combination baseline for completeness and to provide a direct point of comparison with the DRA failure sequences presented in Section 4.5. Whereas DRA failures are characterized by cluster convergence (robots over-committing to a dense target subgroup), the linear baseline exhibits a distinct and complementary failure mode: insufficient persistence, manifesting as frequent target switching and short track dwell intervals.

Figure C.1 presents Sequence L1, a representative linear baseline failure episode. Across the four frames, robots repeatedly abandon established tracks and redirect toward nearby targets, resulting in fragmented tracking coverage. Unlike the DRA failure sequences (F1 and F2), robots in L1 do not converge to a single cluster; instead, each robot independently switches targets at a high frequency, preventing any target from being observed consistently across the episode. This behavior is consistent with the absence of a dedicated persistence critic head in the linear scalarization framework: the persistence gradient is blended with the larger exploration gradient and is suppressed during backpropagation, as discussed in Section 3.8.

Together with the DRA failure analysis in Section 4.5, this sequence illustrates that linear scalarization and decomposed reward training fail in qualitatively different ways: linear scalar-

Sequence L1
Linear
baseline
failure

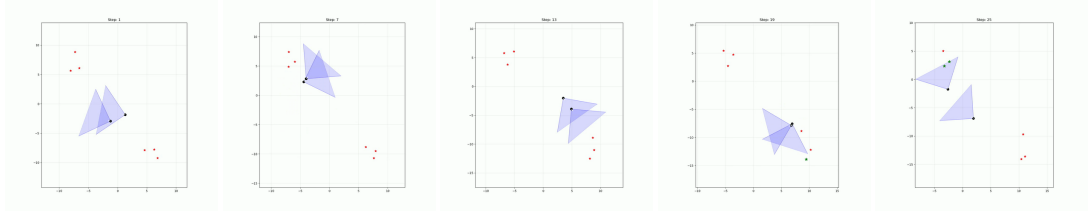


Figure C.1: Linear reward baseline failure Sequence L1. Robots exhibit frequent target switching and short track dwell intervals throughout the episode, contrasting with the cluster-convergence failure mode of the DRA policy. The absence of a dedicated persistence critic head in the linear framework allows the exploration gradient to suppress persistence learning during training.

ization trades away persistence for exploration, while DRA occasionally over-exploits local target clusters at the expense of global coverage. Both failure modes point to the same underlying design challenge—balancing local information gain against global team-level coverage—which is identified as a primary direction for future work in Section 5.3.